

Stat 3113 Exam 1 - Practice Problems Set 2 - Key

1. d. (because $P(A|B) = P(A)$).

2. b.

3. c.

4.a. $S = \{H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6\}$

b. $A = \{T1, T2, T3, T4, T5, T6\}$ $P(A) = \frac{6}{12} = \frac{1}{2}$

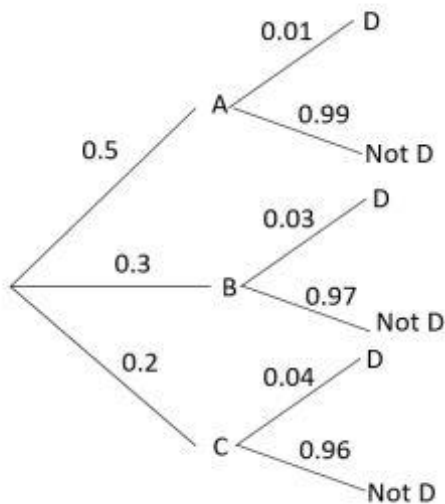
c. $B = \emptyset$ $P(B) = 0$

d. $C = \{H4, H5, H6, T4, T5, T6\}$ $P(C) = \frac{6}{12} = \frac{1}{2}$

e. Not mutually exclusive because $A \cap C = \emptyset$;

They are independent because $P(A|C) = \frac{P(A \cap C)}{P(C)} = \frac{\frac{3}{12}}{\frac{1}{2}} = \frac{1}{2} = P(A)$

5.



$$P(B|D) = \frac{P(B \cap D)}{P(D)} = \frac{(0.3)(0.03)}{(0.5)(0.01) + (0.3)(0.03) + (0.2)(0.04)} = \frac{0.009}{0.022} = 0.41$$

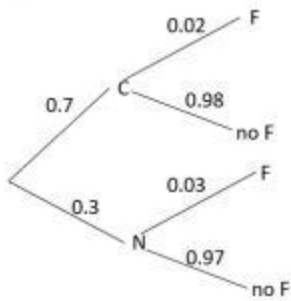
6.

a. 0.4

b. $P(\text{decaf} | \text{small}) = \frac{P(\text{decaf} \cap \text{small})}{P(\text{small})} = \frac{0.2}{0.34} = 0.59$

c. $P(\text{small} | \text{decaf}) = \frac{P(\text{small} \cap \text{decaf})}{P(\text{decaf})} = \frac{0.2}{0.4} = 0.5$

7.



a. $P(F) = (0.7)(0.02) + (0.3)(0.03) = 0.023$

b. $P(C|F) = \frac{P(C \cap F)}{P(F)} = \frac{(0.7)(0.02)}{0.023} = 0.61$

8. experiment – the educator controls the data collection by placing the students into treatment groups.

9. c. – If you look at the boxplot, the maximum, Q3, and median are all tightly grouped together. Then, it begins to stretch out to Q1 and the minimum. This indicates left skew. The histogram would look like this:



10.

- a) 0.405
- b) 0.85

11.

- a) 0.84
- b) 0.5

12.

- a) Yes, only one event can be held at a time.
- b) 0.34

13.

- a) Yes, a person cannot both know how to play and not know how to play a musical instrument.
- b) No, a person can know how to play both the clarinet and the guitar.

14.

- a) A student knows how to play clarinet or guitar or both.
- b) A student does not know how to play guitar.
- c) A student does not know how to play clarinet but does know how to play guitar.

15.

- a) $S = \{1, 2, 31, 32, 41, 42, 341, 342, 431, 432\}$
- b) $A = \{1, 2\}$
- c) $B = \{341, 342, 431, 432\}$